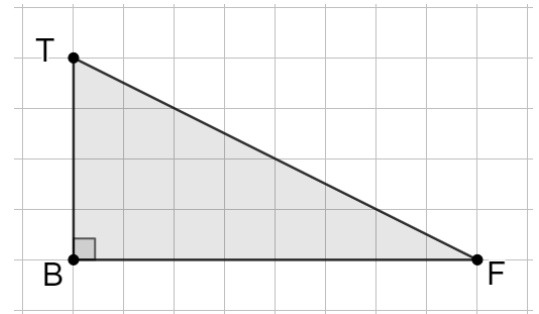


Triangle TBF is shown on a grid.

Construct $\triangle HKM$ so that $\overline{TB} \cong \overline{KM}$, $\angle TBF \cong \angle HMK$, and $\overline{FB} \cong \overline{HM}$ using the given steps, questions 1 through 4.



Steps	Construction
1. Draw $\overline{KM}$ on the grid so that $KM = TB$.	
2. Draw $\angle HMK$ on the grid so that $m\angle HMK = m\angle TBF$.	
3. Draw $\overline{HM}$ on the grid so that $HM = FB$.	
4. Draw $\overline{KH}$.	

5. Describe how to use geometric tools to determine if $\overline{TF} \cong \overline{KH}$, $\angle BFT \cong \angle MHK$, and $\angle FTB \cong \angle HKM$.

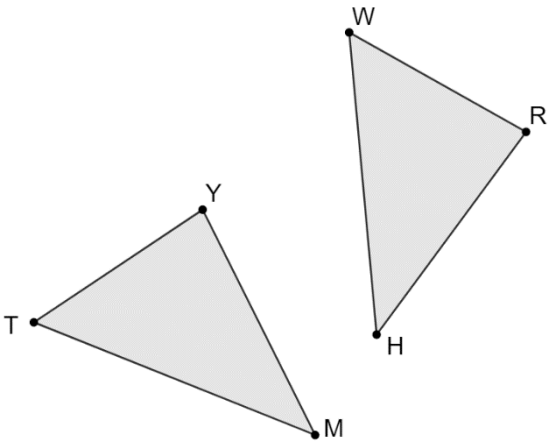
A ruler can be used to compare the measurement of $\overline{TF}$ and $\overline{KH}$. A protractor can be used to measure $\angle BFT$, $\angle MHK$, $\angle FTB$, and $\angle HKM$.

6. Since $KM = TB$, $m\angle HMK = m\angle TBF$, and $HM = FB$, then by the definition of congruence $\overline{TB} \cong \overline{KM}$, $\angle TBF \cong \angle HMK$, and $\overline{FB} \cong \overline{HM}$. Which congruence postulate or theorem does the construction illustrate?

Side-angle-side congruence postulate

Triangles YMT and WRH are shown.

Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.



	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
7.	$\overline{TY} \cong \overline{WR}, \overline{YM} \cong \overline{HR}, \overline{TM} \cong \overline{WH}$	Yes	SSS
8.	$\overline{YM} \cong \overline{HR}, \angle YTM \cong \angle RWH, \overline{TM} \cong \overline{WH}$	No	
9.	$\angle YTM \cong \angle RWH, \overline{TY} \cong \overline{WR}, \overline{TM} \cong \overline{WH}$	Yes	SAS
10.	$\angle MYT \cong \angle HRW, \overline{YM} \cong \overline{HR}, \angle YMT \cong \angle RHW$	Yes	ASA